

ROLES & RESPONSIBILITY OF TEAM MEMBERS [Form – 2]

PROJECT ID: SKIT/IOT/2023-2027/P-04

BRANCH: Computer Science & Engineering (IoT)

SECTION: A

TITLE OF PROJECT: AdityaScan: Multi-Modal Solar Flare Forecasting using Aditya-L1 Telemetry

TEAM LEAD Mahansh Gaur		SPRINT(s) NAME Sprints 1 - 6 (Core AI/ML & MLOps Pipeline)	
USER STORY	START DATE	END DATE	DETAILS OF TASK COMPLETED UNDER THE USER STORY
US-ML-01: Multi-Modal Time-Series Baseline Architecture	10-08-2026	18-09-2026	Implemented bi-directional GRU & LSTM time-series backbones; integrated cross-modal self-attention fusion; established weighted focal loss for flare class imbalance; built baseline benchmarking suite.
US-ML-02: Solar Cycle 25 Dataset & GPU Training Pipeline	21-09-2026	23-10-2026	Engineered chronological train/val/test split generator; resolved extreme class imbalance (X/M/C flares vs quiet Sun); built distributed PyTorch DataLoader chunking for Kaggle/AWS GPU clusters.
US-ML-03: Spatial GNN & Cross-Attention Feature Fusion	26-10-2026	27-11-2026	Implemented PyTorch Geometric Graph Neural Network (GNN) for active region magnetic topology; coupled cross-attention pooling layer; achieved validation TSS > 0.82 and AUC-ROC > 0.88 across 24h horizons.
US-ML-04: ONNX Runtime Engine & Low-Latency Optimization	30-11-2026	31-12-2026	Exported PyTorch fusion models to ONNX Opset 17; applied dynamic INT8 weight quantization; reduced inference latency from 180ms to 42ms; deployed automated health checks & keep-alive monitor.
US-ML-05: Continuous Learning & Checkpoint Serialization	04-01-2027	05-02-2027	Built automated incremental weight updater (run_balanced_update.sh); integrated model registry checkpoint serialization; configured automated regression testing on new NOAA/ISRO telemetry.
US-ML-06: SOTA Evaluation, TRD & Final Project Defense	08-02-2027	15-03-2027	Conducted comprehensive comparative audit against NOAA SWPC & NASA baseline models; generated final evaluation metrics report (TSS=0.887, Brier=0.089); authored final project report & presentation.

MEMBER1 Sanskriti Raj		SPRINT(s) NAME Sprints 1 - 6 (Aditya-L1 Ingestion & Physics)	
USER STORY	START DATE	END DATE	DETAILS OF TASK COMPLETED UNDER THE USER STORY
US-ING-01: Aditya-L1 Multi-Payload Ingestion Pipeline	10-08-2026	18-09-2026	Developed ISRO PRADAN batch downloader; implemented Level-1 FITS parser for SoLEXS soft X-ray spectrometer; calibrated HELIOS hard X-ray flux; configured bad packet telemetry rejection.
US-ING-02: NOAA GOES & SDO-SHARP Live Sync Poller	21-09-2026	23-10-2026	Built async NOAA GOES-16/18 XRS primary/secondary fetcher; implemented SDO JSOC SHARP cylindrical equal-area (CEA) magnetic vector patch downloader with automated caching.
US-PHY-03: Neupert Proxy & Wavelet QPP Pulsation Detection	26-10-2026	27-11-2026	Implemented Neupert effect derivative proxy (dI/dt proportional to HXR); built Continuous Morlet Wavelet Transform (CWT) to detect 10-300s quasi-periodic pulsations in solar flux.
US-PHY-04: In-Situ Solar Wind & Plasma Vector Calibration	30-11-2026	31-12-2026	Built Aditya-L1 ASPEX/SWIS proton/alpha energy spectrum parser (50 channels); implemented MAG triaxial fluxgate vector baseline correction and solar wind plasma flow calibrator.
US-CAL-05: Conformal Uncertainty & Temperature Scaling	04-01-2027	05-02-2027	Developed Monte Carlo Dropout (50 passes) and Platt temperature scaling (reduced ECE to <3%); implemented conformal prediction quantile calibrators guaranteeing 90% coverage.
US-PHY-06: Scientific Cross-Validation & Technical Report	08-02-2027	15-03-2027	Authored 28-page comprehensive AdityaScan Technical Report; verified physical correlation between SoLEXS flare peaks and ASPEX plasma arrivals; prepared defense exhibits.

MEMBER2 <u>Somya Roy</u>		SPRINT(s) NAME <u>Sprints 1 - 6 (Mission Control UI & DevOps)</u>	
USER STORY	START DATE	END DATE	DETAILS OF TASK COMPLETED UNDER THE USER STORY
US-FS-01: FastAPI Server & Glassmorphic Mission UI	10-08-2026	18-09-2026	Initialized FastAPI server with SQLite/PostgreSQL schema; built React 18 / Vite application with Tailwind CSS and TanStack Router; configured responsive dark-mode glassmorphic theme.
US-FS-02: Real-Time Telemetry & Lightcurve Visualization	21-09-2026	23-10-2026	Integrated high-performance Recharts components for live X-ray and magnetometer curves; built active solar region status cards (AR13664, AR13668); added TanStack Query hooks.
US-FS-03: Multi-Horizon Forecast Probability Gauges	26-10-2026	27-11-2026	Built SVG forecast probability gauges displaying 30m, 1h, 3h, 6h, and 24h risk; visualized conformal uncertainty bands [p_low -> p_high]; added flare classification badges.
US-FS-04: Sub-Second WebSocket Live Telemetry Engine	30-11-2026	31-12-2026	Engineered /ws/live WebSocket streaming endpoint with connection manager and broadcast queue; built automated reconnect hook in React; reduced UI refresh latency to <500ms.
US-FS-05: Constellation Protection & Safe-Mode Guidance	04-01-2027	05-02-2027	Added NavIC & GSAT orbital constellation radiation alert banners; built interactive satellite safe-mode procedure checklist modal; implemented May 2024 storm demo scenario injector.
US-OPS-06: Production Containerization, CI/CD & Deploy	08-02-2027	15-03-2027	Configured multi-stage Docker build and docker-compose orchestration; deployed web frontend to Cloudflare Pages and backend to Render; set up GitHub Actions CI/CD workflows.

MEMBER3 <u>N/A (Three-Member Team)</u>		SPRINT(s) NAME <u>Not Applicable (3-Member Project)</u>	
USER STORY	START DATE	END DATE	DETAILS OF TASK COMPLETED UNDER THE USER STORY
Sprint 1: [Not Applicable]	10-08-2026	18-09-2026	Three-Member Final Year Project Group: Mahansh Gaur (Lead), Sanskriti Raj (Member 1), Somya Roy (Member 2).
Sprint 2: [Not Applicable]	21-09-2026	23-10-2026	Three-Member Final Year Project Group: Mahansh Gaur (Lead), Sanskriti Raj (Member 1), Somya Roy (Member 2).
Sprint 3: [Not Applicable]	26-10-2026	27-11-2026	Three-Member Final Year Project Group: Mahansh Gaur (Lead), Sanskriti Raj (Member 1), Somya Roy (Member 2).
Sprint 4: [Not Applicable]	30-11-2026	31-12-2026	Three-Member Final Year Project Group: Mahansh Gaur (Lead), Sanskriti Raj (Member 1), Somya Roy (Member 2).
Sprint 5: [Not Applicable]	04-01-2027	05-02-2027	Three-Member Final Year Project Group: Mahansh Gaur (Lead), Sanskriti Raj (Member 1), Somya Roy (Member 2).
Sprint 6: [Not Applicable]	08-02-2027	15-03-2027	Three-Member Final Year Project Group: Mahansh Gaur (Lead), Sanskriti Raj (Member 1), Somya Roy (Member 2).

Verified & Approved By:
(MENTOR’S NAME & SIGNATURE)

Project Mentor